

The Open University of Sri Lanka
Faculty of Engineering Technology
Department of Electrical and Computer Engineering



Study Programme	: Bachelor of Software Engineering Honours
Name of the Examination	: Final Examination
Course Code and Title	: EEI6567 Software Architecture and Design
Academic Year	: 2023/2024
Date	: 08/02/2025
Time	: 13.30hrs – 16.30hrs
Duration	: 3 hrs.

General Instructions

1. Read all instructions carefully before answering the questions.
 2. This question paper consists of **2 sections, A and B** and questions in **four (4)** pages.
 3. Answer all questions from **Section A** and answer any **3** questions from **Section B**.
 4. Answer for each question should commence from a new page.
 5. Draw fully labelled diagrams where necessary.
 6. Write answers in clear and concise format appropriately in points/bullet form.
 7. Use blue or black ink pen to answer the questions.
 8. Clearly state your index number in every page of the answer script.
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Section A

- 1) Your company has been asked to develop a software system for 'Automating Store Management in a Hotel System', and you are the architect.

"Managing store inventory and procurement in our hotel is highly manual and inefficient, leading to frequent stockouts of essential items, overstocking of non-essential goods, and inaccurate inventory records. These issues result in increased operational costs, delayed service delivery, and poor guest experience.

How can we implement an automated store management system that optimizes inventory levels, tracks usage in real-time, and integrates with procurement to streamline operations and reduce costs?"

Key Considerations:

- Inventory Tracking: Real-time tracking of stock levels for food, beverages, cleaning supplies, and other essentials.
 - Supplier Integration: Automate purchase orders when stock levels fall below predefined thresholds.
 - User Access and Permissions: Allow different departments (kitchen, housekeeping, etc.) to request and track supplies.
 - Analytics and Reporting: Provide insights into stock usage, costs, and trends to optimize procurement.
- a) Discuss the significance of automating a store management system for hotels, highlighting the benefits and drawbacks of the associated costs. **(5 Marks)**
 - b) As an architect, what do you consider as your responsibilities? **(5 Marks)**
 - c) List down the stakeholders of this system **(5 Marks)**
 - d) Draw a use case diagram for this system **(5 Marks)**
 - e) Identify major software components **(5 Marks)**
 - f) Draw a diagram to depict the high-level logical architecture of this system you are going to design **(10 Marks)**
 - g) Discuss two non-functional requirements that need to be considered in this system and explain why these are important in this system **(5 Marks)**

Section B (Answer any 3 questions from below)

- 2) Microservice architecture and cloud deployment are essential considerations in contemporary software design and delivery. Many architects favor microservices for software development and opt for cloud platforms to manage and maintain the software infrastructure. Please provide detailed responses to the following questions.
- a) What is Monolithic Architecture? Provide its key characteristics. **(6 Marks)**
 - b) Explain the difference between monolithic and microservices architecture. **(4 Marks)**
 - c) Name 4 types of cloud and describe each with its characteristics. **(4 Marks)**
 - d) What are the 3 main common Cloud service models? Describe them with their characteristics. **(6 Marks)**
- 3) Software Evolution and Software Maintenance are critical topics in the software industry, reflecting key trends shaping the modern landscape. These practices ensure software systems remain relevant, efficient, and adaptable to changing requirements and technologies. Please provide detailed responses to the following questions.
- a) What is the difference between Software Maintenance and Software Evolution? **(4 Marks)**
 - b) There are 4 main types of software Maintenance; name and describe each. **(4 Marks)**
 - c) Identify the five primary system artifacts that undergo changes during the software evolution process. **(5 Marks)**
 - d) What is impact analysis? Describe the steps you follow when performing an impact analysis. **(7 Marks)**
- 4) The primary focus of software architecture is to provide effective solutions for both the functional and non-functional requirements of a software system. Software design patterns and quality attributes play critical roles in defining and ensuring the overall quality of the system. Please answer the following questions.
- a) Describe the three primary classifications of software design patterns and provide two examples for each category. **(6 Marks)**
 - b) Explain the importance of software quality attributes in the software development process including the main 3 categories of software quality attributes. Give 2 examples for each category **(8 Marks)**
 - c) Performance is the key software quality attribute that we must consider while designing the system. List down 3 actions that we can take to improve the performance of the software system. **(3 Marks)**
 - d) Software security is one of the important non-functional aspects that we must consider for any software development. As an architect what are the decisions that

you are taking to enhance the security aspect of your system? Explain 3 decisions. (3 Marks)

- 5) Providing solutions for recurring software problems is one of the major responsibilities of a software architect. Please answer the questions below on software architectural concerns.
- a) Describe 4+1 View models and provide examples diagrams (UML or non-UML diagrams) that can be used to illustrate each view model. (6 Marks)
 - b) Explain the advantages and disadvantages of the Agile software development life cycle compared to the traditional Waterfall methodology. (4 Marks)
 - c) Describe the key Scrum ceremonies. (6 Marks)
 - d) List down 4 decisions that you can take during the user interface design to improve the usability of the system. (4 Marks)

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